STATISTICS WORKSHEET-8

1. B
2. B
3. D
4. B
5. B
6. D
7. B
8. A
9. A
10. A
11. A
12. D

13. What is Anova in SPSS?

ANOVA (Analysis of Variance) is a statistical technique used to test the significance of differences between two or more groups or treatments. In SPSS, ANOVA is used to analyze the variation in a continuous dependent variable across two or more categorical independent variables.
SPSS offers several types of ANOVA, including one-way ANOVA, factorial ANOVA, repeated-measures ANOVA, and mixed-design ANOVA. The choice of which ANOVA to use depends on the specific research question and the experimental design.

Overall, ANOVA is a powerful tool in SPSS for analyzing differences between groups or treatments and can be used in a wide range of research contexts, including social sciences, medical research, and engineering.

14. What are the assumptions of Anova?

The assumptions of ANOVA (Analysis of Variance) are:

Independence: The observations in each group must be independent of each other. This means that the value of one observation does not depend on the value of another observation.

Normality: The data within each group should follow a normal distribution. This means that the distribution of data should be bell-shaped.

Homogeneity of variances: The variance of the data within each group should be approximately equal. This means that the spread of the data should be similar for all groups.

15. What is the difference between one way Anova and two way Anova?

One-way ANOVA (Analysis of Variance) and two-way ANOVA are statistical tests used to analyze the differences between the means of two or more groups or factors. The main difference between the two is the number of factors or independent variables being analyzed.

In one-way ANOVA, there is only one factor or independent variable being analyzed. For example, if we want to compare the mean scores of three different groups of students (e.g., group 1, group 2, and group 3) on a particular exam, we would use one-way ANOVA. This test examines whether there is a significant difference between the means of the three groups.

On the other hand, in two-way ANOVA, there are two factors or independent variables being analyzed simultaneously. For example, if we want to compare the mean scores of three different groups of students on a particular exam, but also want to see if the gender of the students affects the scores, we would use two-way ANOVA. This test examines whether there is a significant difference between the means of the three groups and whether the gender of the students has an effect on the scores.

In summary, the main difference between one-way ANOVA and two-way ANOVA is the number of factors or independent variables being analyzed. One-way ANOVA analyzes the differences between groups based on one factor, while two-way ANOVA analyzes the differences between groups based on two factors simultaneously.